

RESEARCH INTERESTS

Theoretical Cryptography, Post-Quantum Cryptography, Provable Security, Memory-Hard Computation, Differential Privacy, and Combinatorial Methods in Security

ACADEMIC APPOINTMENTS

Jul. 2025 – Present	Postdoctoral Researcher <i>Postdoc Supervisor:</i> Douglas Stebila	University of Waterloo Department of Combinatorics and Optimization
Jul. 2024 – Jun. 2025	Postdoctoral Researcher <i>Postdoc Supervisor:</i> Jeremiah Blocki	Purdue University Department of Computer Science

EDUCATION

Aug. 2017 – May 2024	Ph.D. in Computer Science <i>Dissertation:</i> Applications of Combinatorial Graph Theory to the Classical and Post-Quantum Security Analysis of Memory-Hard Functions and Proofs of Sequential Work <i>Advisor:</i> Jeremiah Blocki	Purdue University
Sep. 2010 – Feb. 2013	M.S. in Mathematics <i>Thesis:</i> Reinitializing Techniques in Level Set Method <i>Advisor:</i> Myungjoo Kang <i>Additional doctoral study in Mathematics, Mar. 2013 – Dec. 2013, prior to mandatory military service.</i>	Seoul National University
Mar. 2005 – Feb. 2010	B.S. in Mathematics Graduated <i>magna cum laude</i> , Recipient of the <i>Presidential Science Scholarship</i>	POSTECH (Pohang University of Science and Technology)

PUBLICATIONS AND PREPRINTS

(Note: Authors are listed in alphabetical order by their last name.)

Submitted Manuscripts

1. Sparse Depth-Robust Graphs with Concrete Improved Lower Bounds

Jeremiah Blocki, Jong Chan Lee, Seunghoon Lee, Peiyuan Liu, and Ling Ren

2. On the Multi-User Security of CSI-FiSh with Tight Reductions

Seunghoon Lee, Maher Mamah, and Bruno Sterner

Cryptology ePrint Archive, 2026.

3. On the Preprocessing Security of Nonzero Schnorr Signatures with Adaptive Corruptions

Jeremiah Blocki and Seunghoon Lee

Cryptology ePrint Archive, 2025.

Publications

4. On the Additive Sensitivity of LZ77 Under Consecutive Edits

Jeremiah Blocki and Seunghoon Lee

To appear in Theory of Cryptography Conference (TCC 2026)

5. A Tight Lower Bound on the TdScrypt Trapdoor Memory-Hard Function

Jeremiah Blocki and Seunghoon Lee

In IACR Communications in Cryptology, Volume 2, Issue 3 (2025)

6. Differentially Private Compression and the Sensitivity of LZ77

Jeremiah Blocki, Seunghoon Lee, and Brayan Sebastián Yepes Garcia

In Theory of Cryptography Conference (TCC 2025)

7. The Impact of Reversibility on Parallel Pebbling

Jeremiah Blocki, Blake Holman, and Seunghoon Lee

In Advances in Cryptology – EUROCRYPT 2025

8. **Differentially Private L_2 -Heavy Hitters in the Sliding Window Model**
Jeremiah Blocki, Seunghoon Lee, Tamalika Mukherjee, and Samson Zhou
In The Eleventh International Conference on Learning Representations (ICLR 2023)
9. **The Parallel Reversible Pebbling Game: Analyzing the Post-Quantum Security of iMHFs**
Jeremiah Blocki, Blake Holman, and Seunghoon Lee
In Theory of Cryptography Conference (TCC 2022)
10. **On the Multi-User Security of Short Schnorr Signatures with Preprocessing**
Jeremiah Blocki and Seunghoon Lee
In Advances in Cryptology – EUROCRYPT 2022
11. **On Explicit Constructions of Extremely Depth Robust Graphs**
Jeremiah Blocki, Mike Cinkoske, Seunghoon Lee, and Jin Young Son
In 39th International Symposium on Theoretical Aspects of Computer Science (STACS 2022)
12. **On the Security of Proofs of Sequential Work in a Post-Quantum World**
Jeremiah Blocki, Seunghoon Lee, and Samson Zhou
In 2nd Conference on Information-Theoretic Cryptography (ITC 2021)
13. **Approximating Cumulative Pebbling Cost is Unique Games Hard**
Jeremiah Blocki, Seunghoon Lee, and Samson Zhou
In 11th Innovations in Theoretical Computer Science Conference (ITCS 2020)
14. **Data-Independent Memory Hard Functions: New Attacks and Stronger Constructions**
Jeremiah Blocki, Benjamin Harsha, Siteng Kang, Seunghoon Lee, Lu Xing, and Samson Zhou
In Advances in Cryptology – CRYPTO 2019

In Preparation

- **Hybrid AKEM: Constructions and Security Properties**
Axel Durbet, Seunghoon Lee, Reihaneh Safavi-Naini, and Douglas Stebila
- **Towards Generic Isogeny Model**
Seunghoon Lee, Maher Mamah, Youcef Mokrani, and Bruno Sterner
- **On the Preprocessing Security of TLS Handshake**
Jeremiah Blocki, Seunghoon Lee, and Douglas Stebila

Manuscript

- **A Short Note on Improved Logic Circuits in a Hexagonal Minesweeper**
Seunghoon Lee

TEACHING EXPERIENCE

Instructor of Record

- **MATH 137: Calculus 1 for Honours Mathematics**, University of Waterloo (Fall 2026)
- **MATH 138 Online: Calculus 2 for Honours Mathematics**, University of Waterloo (Spring 2026)

Teaching Assistant and Course Development

- **CS 58000-DEV: Algorithm Design, Analysis, and Implementation — Online Course Development**, Purdue University (Fall 2021)
- **CS 51500: Numerical Linear Algebra**, Purdue University (Fall 2018)
- **CS 25100: Data Structures and Algorithms**, Purdue University (Fall 2017, Spring 2018)
- **300.204: Differential Equations**, Seoul National University (Spring/Fall 2013)
- **033.002: Calculus 2**, Seoul National University (Fall 2010, Fall 2013)
- **033.001: Calculus 1**, Seoul National University (Spring 2013)
- **033.004: Honor Calculus and Practice 2**, Seoul National University (Fall 2012)
- **046.001: Mathematics in Civilization**, Seoul National University (Spring/Fall 2011, Spring 2012)
Received Outstanding TA Award (Spring 2012)

TEACHING DEVELOPMENT

- **Fundamentals of University Teaching**, Centre for Teaching Excellence, University of Waterloo — *in progress; expected completion Oct. 2026*

MENTORING AND OUTREACH

Undergraduate Mentoring

Spring/Fall 2024 Brayan Sebastián Yepes Garcia Purdue University & Universidad Nacional de Colombia
Topic: Differentially Private Compression and the Sensitivity of LZ77
Resulted in a coauthored publication at TCC 2025.

K-12 Outreach

Fall 2013 Research and Education Program Sejong Science High School
Topic: Finite Difference Method in Numerical Analysis

TALKS AND POSTER PRESENTATIONS

Future Presentations

Nov. 2026 On the Additive Sensitivity of LZ77 Under Consecutive Edits TCC 2026

Conference Presentations

Dec. 2025 Differentially Private Compression and the Sensitivity of LZ77 TCC 2025
Nov. 2022 The Parallel Reversible Pebbling Game: Analyzing the Post-Quantum Security of iMHFs TCC 2022
Mar. 2022 On Explicit Constructions of Extremely Depth Robust Graphs STACS 2022
Jul. 2021 On the Security of Proofs of Sequential Work in a Post-Quantum World ITC 2021
Jan. 2020 Approximating Cumulative Pebbling Cost is Unique Games Hard ITCS 2020

Invited and Seminar Talks

Jul. 2026 On the Multi-User Security of Short Schnorr Signatures with Preprocessing University of Calgary
Mar. 2026 Parallel Reversible Pebbling: Time-Space Tradeoffs on DAGs
(with Cryptographic Motivation) Tutte Colloquium
Feb. 2026 Differentially Private Compression and the Sensitivity of LZ77 Act2Qrypt Virtual Seminar
Dec. 2023 Multi-User Security of Short Schnorr Signatures with Preprocessing Purdue Crypto Reading Group
Nov. 2019 Approximating Cumulative Pebbling Cost is Unique Games Hard Purdue Crypto Reading Group

Posters

Mar. 2022 On the Multi-User Security of Short Schnorr Signatures with Preprocessing CERIAS Symposium 2022
Jan. 2020 Approximating Cumulative Pebbling Cost is Unique Games Hard ITCS 2020
Apr. 2019 On the Security of Short Schnorr Signatures Midwest Security Workshop 7
Apr. 2019 On the Security of Short Schnorr Signatures CERIAS Symposium 2019

RESEARCH VISITS

Jul.-Aug. 2026 Research Visit University of Calgary
Collaborative research in theoretical and post-quantum cryptography; hosted by Prof. Reihaneh Safavi-Naini.

PROFESSIONAL AND ACADEMIC SERVICE

Conference and Journal External Reviewing

CCS 2019, NDSS 2020, CT-RSA 2020, ITC 2020, CRYPTO 2020, TCC 2020, CRYPTO 2021, ITCS 2022, FC 2022, ITC 2022, CRYPTO 2022, SYNASC 2022, IEEE S&P 2023, EUROCRYPT 2023, IEEE S&P 2024, EUROCRYPT 2024, ITC 2024, ESA 2024, QIP 2025, IEEE S&P 2025, RANDOM 2025, TCC 2025, Communications in Cryptology (Volume 2, Issue 4), DCC 2026, ACNS 2026, and TCC 2026.

University of Waterloo Crypto Reading Group		
Spring 2026	Code-Based Cryptography	Co-organizer
Winter 2026	Advanced Public-Key Encryption	Co-organizer
Fall 2025	Post-Quantum Digital Signatures	Co-organizer

ADDITIONAL PROFESSIONAL EXPERIENCE		
Dec. 2013 – Dec. 2016	Senior Researcher (mandatory military service)	Security Management Institute

HONORS, FELLOWSHIPS, AND AWARDS		
April 2025	Postdoctoral Mentor Award Nominee for the Office of the Vice Provost for Graduate Students and Postdoctoral Scholars	Purdue University
Aug. 2023 – May 2024	Bilsland Dissertation Fellowship	Purdue University
Spring 2012	Outstanding Teaching Assistant Award	Seoul National University
Sep. 2010 – Dec. 2013	Brain Korea 21 Scholarship	National Research Foundation of Korea
Mar. 2005 – Feb. 2010	Presidential Science Scholarship	Korea Student Aid Foundation